## 1 Adaptive grouping

from the task and the agent's thoughts only

**Task**

Export the slides as a PDF file

plus agent thoughts

**Group 1** Open file — step 1, step 2, step 3

**Group 2** Edit slides — step 4, step 5, step 6

**Group 3** Export PDF — step 7, step 8, step 9

## 2 Two-level scoring

score inside a group, refine across groups, then combine

**within one group** — Group 2 · Edit slides

**Reference** — what this group is trying to do · used when scoring steps and the group

target state | constraints | preferred process

**Action-level** — each column is one step

| step 4 | step 5 | step 6 |
|---|---|---|
| Done? executed | Done? executed | Done? partial |
| Helpful? useful | Helpful? milestone | Helpful? redundant |
| screen change · later steps in this group | screen change · later steps in this group | screen change · later steps in this group |
| step score | step score | step score |

**Group-level** — the same three steps, read as one subgoal

subgoal completed? does it help the overall task? → group score: Edit slides

**across all groups**

**Refinement**

later groups can change an earlier group's role

first pass (one score per group)

| Open file | Edit slides | Export PDF |
|---|---|---|
| high | medium | low |

look at every group together plus the final task outcome

after seeing what followed

| Open file | Edit slides | Export PDF |
|---|---|---|
| high | low | low |

**Combine** — each step keeps its step score; the group contributes one refined score

step score × refined group score = process reward

## 3 PR-GRPO

finer credit than outcome-only GRPO

**learning signal = outcome + process**

- **outcome** — same for every step
- **process** — varies across steps